Diego Miguel Reynoso Ramos
• Aerospace Engineering at UCLA •

323-362-3826 | diegomrr@g.ucla.edu | Los Angeles, CA | https://www.diegoreynoso.com/

EDUCATION

University of California, Los Angeles (UCLA) — Los Angeles, CA
Pursuing B.S., Aerospace Engineering | ***GPA 3.8*** — ***Expected June 2027***

East Los Angeles College (ELAC) — Monterey Park, CA
AS., Mathematics — *August 2022 – August 2024*

TECHNICAL EXPERIENCE

UCLA Energy and Propulsion Laboratory — Los Angeles, CA
Undergraduate Researcher, — *May 2025 - Present*

- Collected and analyzed shear-layer velocity data via hot-wire anemometry, using **MATLAB** to evaluate turbulence and jet interactions for airbreathing engine applications.
- Designed and modeled custom calibration and measurement components in **Fusion 360** to improve the accuracy and efficiency of the experimental setup.
- Presented research findings at a university-wide poster symposium attended by 50+ undergraduate researchers.

Design Build Fly (DBF) — Los Angeles, CA
CAD, Aerodynamics, and Structures Subteams member. — *May 2025 - Present*

- Performed **ANSYS FEM** simulations on components and executed static/strain tests to validate simulation data.
- Ran **CFD** studies with **COMSOL/SIMSCALE** to characterize flow and compute aerodynamic loads; validated results against wind-tunnel/field data for d
- Modeled aircraft components in **SolidWorks** for assembly and **fabrication**.

Rocket Project — Los Angeles, CA
Team Lead — *August 2024 - November 2024*

- Led a 3-person team in the design cycle of a G-class rocket, coordinating **fabrication** and **integration**.
- Designed and manufactured flight-tested nose cones, fins, and body tubes in **SolidWorks**, fiberglass/epoxy, and **carbon fiber layups**.

Mars Rover Project, 2024 UCLA HAcK Competition — Los Angeles, CA
Manufacturing Lead — *July 2024 - July 2024*

- Developed a remotely controlled rover equipped with a working camera, along with humidity, temperature, and ultrasonic sensors, all accessible through a custom-built website with **HTML/CSS**.
- Designed and fabricated chassis, claw, and arm mechanisms using 3D printing and laser cutting.
- Coordinated mechanical integration with electronics and software teams.

WORK EXPERIENCE

MESA at East Los Angeles — Monterey Park, CA
Lead Mathematics Tutor — *March 2023 - September 2024*

- Tutored university level Mathematics, Chemistry, and Physics, collaborating with staff to aid student success.
- Guided group tutoring sessions for up to 8 students in Physics, facilitating collaborative problem-solving.

McDonald's — Huntington Park, CA
Shift Lead — *August 2022 - July 2024*

- Supervised 10+ employees per shift; trained new staff on workflow, and oversaw daily transactions (~$15K).
- Resolved customer concerns efficiently as well as staff conflicts while maintaining service standards, contributing to high satisfaction scores and smooth shift operations.

RELEVANT COURSEWORKS

MATLAB Programming, Introduction to Astronautics, Analysis of Aerospace Structures, Flight Mechanics, Stability, and Control of Aircraft, Thermodynamics, Transport Phenomena, Fluid Mechanics, Aerodynamics, Circuits, Materials Science, Statics, Strength of Materials, Dynamics.

SKILLS

Skills: CAD: SolidWorks/Onshape/Fusion, CFD: COMSOL/SimScale, FEM: Ansys, Arduino, HTML/CSS, 3D Printing, Laser Cutting, Machining, Fiberglass Fabrication, Soldering, Mechanical Design, Prototyping, and Teamwork.